

Box-iLQR: Constraint-Aware Feedback via Finite-Barrier Trajectory Optimization

Abstract—Iterative Linear Quadratic Regulator (iLQR) and Differential Dynamic Programming (DDP) are attractive trajectory-optimization methods because their backward recursions provide both nominal trajectories and local time-varying feedback laws. Hard state and input constraints, however, can drive solutions toward constraint boundaries where little corrective authority remains during closed-loop execution.

This paper presents *Box-iLQR*, a log-barrier iLQR method for box-constrained states and controls, with emphasis on the resulting feedback policy. Barrier gradients bias nominal trajectories away from constraint boundaries, while barrier curvature enters the Riccati recursion that determines the feedback gains. We show that control-barrier curvature smoothly attenuates feedback as an actuator approaches saturation, while state-barrier curvature propagates backward through the value function to shape feedback before future state constraints are encountered. This motivates retaining a finite barrier rather than always driving the barrier parameter toward zero, thereby preserving state and actuator margin. Comparisons with CL-DDP and ALTRO are presented on pendulum, cart-pole, Acrobot, and car-parking problems. Monte Carlo experiments demonstrate improved closed-loop behavior with finite barriers, while a six-input MuJoCo fish example demonstrates data-driven Box-iLQR using dynamics Jacobians estimated from simulator rollouts. Box-iLQR achieves convergence comparable to, and in some cases better than, the baseline methods. Moreover, the feedback policy obtained with a finite barrier ensures constraint satisfaction even under execution uncertainty.

I. INTRODUCTION

Trajectory optimization is a fundamental tool for motion planning and control of nonlinear robotic systems. Among shooting-based methods, Differential Dynamic Programming (DDP) and the iterative Linear Quadratic Regulator (iLQR) are particularly attractive because they exploit the temporal structure of the optimal-control problem and produce a time-varying local feedback law together with the optimized nominal trajectory [1], [2], [3]. Modern robotics software and algorithms such as Crocoddyl further demonstrate the practical utility of Riccati-based optimal-control methods for complex nonlinear systems [4].

Among Riccati-based trajectory-optimization methods, we focus on iLQR rather than full DDP. Recent Sequential Quadratic Programming (SQP) interpretations show that iLQR provides a consistent quadratic approximation of the discrete optimal-control problem, while the additional dynamics-Hessian terms in DDP can produce indefinite local models and poor cost predictions away from a solution [5], [6]. This makes iLQR particularly suitable for the barrier-continuation setting considered here, where updates of the barrier parameters repeatedly change the local objective and may move the optimization away from its current minimizer.

The feedback returned by iLQR is not merely a numerical byproduct. A related line of work views nonlinear control and reinforcement learning as the search for a nominal maneuver together with a local feedback controller, rather than as the direct search for a high-dimensional global policy [7], [8], [3]. In this view, trajectory optimization determines the operating trajectory, while local perturbation dynamics determine the feedback correction around it.

The central difficulty is that practical systems are constrained. Actuators saturate, joint or workspace coordinates are bounded, and safety requirements can impose explicit limits on states. CL-DDP incorporates box-constrained controls through stagewise constrained quadratic subproblems [9]. Feasibility-driven DDP methods improve globalization and initialization behavior [10], while hybrid and projection methods offer alternative constrained formulations [11], [12], [13]. ALTRO combines an augmented-Lagrangian formulation with structure-exploiting trajectory optimization to address general nonlinear constraints [14]. Interior-point methods provide another important class of constrained optimization techniques, including structured optimal-control solvers and interior-point variants of DDP [15], [16], [17], [18]. A key distinction of the primal log-barrier formulation used in Box-iLQR is that every accepted iterate remains strictly inside the feasible region. In contrast, augmented-Lagrangian methods may pass through infeasible intermediate trajectories while progressively reducing the constraint residual. Primal-dual interior-point methods likewise do not require strict primal feasibility at every intermediate iteration [17], [16], [18]. Box-iLQR, therefore, preserves strict feasibility throughout the accepted optimization sequence, at the cost of requiring a strictly feasible initialization. Moreover, Box-iLQR offers a critical advantage for feedback-oriented deployment: the log-barrier curvature directly shapes the Riccati gains, ensuring closed-loop system satisfies the constraints under uncertainty.

Pure primal barrier methods, however, introduce an important numerical issue. Wright showed that after reducing a barrier parameter, a Newton step can become strongly infeasible even when computed from an interior iterate [19]. Classical nonlinear optimization therefore combines barrier models with globalization mechanisms such as line search or trust regions [20]. Similar behavior is relevant to trajectory optimization because a control perturbation at one time instant affects all downstream states.

This paper takes a complementary viewpoint: if the output of trajectory optimization will be used directly as a *feedback controller*, driving the barrier parameter arbitrarily close to

zero may not be desirable. At a hard-constrained optimum, a control can lie almost exactly at its saturation limit, leaving little authority for disturbance rejection in one direction. Likewise, a nominal state that approaches a path-constraint boundary has little geometric margin against execution uncertainty. Retaining a finite logarithmic barrier intentionally shifts the nominal solution into the feasible interior and simultaneously modifies the local curvature from which the feedback matrix is computed.

The principal contributions of this work are:

- A finite-barrier Box-iLQR formulation that handles simultaneous state and control box constraints while preserving the standard local feedback policy parametrization of iLQR/DDP leading to the structured backward-forward search.
- A characterization of how barrier curvature shapes the feedback gains. As a constrained control approaches its bound, the corresponding row of the feedback matrix tends to zero. On the other hand, the state-barrier curvature propagates backward and modifies the feedback before the state constraint is reached thereby allowing for constraint satisfaction even in the presence of uncertainty.
- A finite-barrier stopping rule for feedback deployment. Instead of driving the barrier parameter to zero, the optimization may terminate at a positive barrier value to retain state and actuator margins for closed-loop execution that ensures constraint satisfaction.
- Numerical validation against CL-DDP and ALTRO, together with a data-driven MuJoCo example in which the local dynamics are identified from simulator rollouts. Experiments show that Box-iLQR is competitive/ better than CL-DDP and ALTRO on nominal constrained optimal control, while the finite-barrier feedback reduces observed state-constraint violations to zero in problems under uncertainty, which the competing methods cannot accomplish.

II. PROBLEM FORMULATION

Consider the discrete-time finite-horizon optimal-control problem

$$\min_{\{u_k\}_{k=0}^{N-1}} J = \phi(x_N) + \sum_{k=0}^{N-1} \ell(x_k, u_k), \quad (1a)$$

$$\text{s.t. } x_{k+1} = f(x_k, u_k), \quad k = 0, \dots, N-1, \quad (1b)$$

$$\underline{x}_i < x_{k,i} < \bar{x}_i, \quad i \in \mathcal{I}_x, \quad (1c)$$

$$\underline{u}_j < u_{k,j} < \bar{u}_j, \quad j \in \mathcal{I}_u. \quad (1d)$$

Here $x_k \in \mathbb{R}^n$, $u_k \in \mathbb{R}^m$, and x_0 is fixed. Only selected state or control components need to be constrained. We assume that f is continuously differentiable and that the nominal running and terminal costs are twice continuously differentiable over the region visited by the optimizer.

A. Continuous- and discrete-time state constraints

If the original plant is continuous time,

$$\dot{x} = f_c(x, u), \quad (2)$$

and $u(t) = u_k$ on $I_k = [t_k, t_{k+1}]$, let $F_{\Delta t}$ denote the exact zero-order-hold flow map. Then

$$x_{k+1} = F_{\Delta t}(x_k, u_k) \quad (3)$$

is an exact sampled representation of the continuous dynamics.

A zero order control hold means the control is not violated for continuous dynamics as well. Although for states, we only have endpoint feasibility as we convert continuous dynamics to discrete. Endpoint feasibility alone does not generally imply continuous-time path feasibility. For a constrained state component x_i , assume additionally that $x_i(t)$ is monotonic on each sampling interval, I_k . Under this explicit assumption,

$$\underline{x}_i \leq x_{k,i}, x_{k+1,i} \leq \bar{x}_i \iff \underline{x}_i \leq x_i(t) \leq \bar{x}_i, \quad t \in I_k. \quad (4)$$

With numerical integration, (4) holds only up to the integration error. This qualification is important whenever discrete nodes are used to represent continuous-time state constraints [21].

A continuous-time performance index in Lagrange form can be readily approximated by the discrete cost in (1) using a numerical transcription scheme such as Euler or Runge–Kutta integration.

B. Log-barrier subproblem

For state-barrier parameters $\mu_i > 0$ and control-barrier parameters $\sigma_j > 0$, define

$$b_x(x_k; \mu) = - \sum_{i \in \mathcal{I}_x} \mu_i [\log(x_{k,i} - \underline{x}_i) + \log(\bar{x}_i - x_{k,i})], \quad (5)$$

$$b_u(u_k; \sigma) = - \sum_{j \in \mathcal{I}_u} \sigma_j [\log(u_{k,j} - \underline{u}_j) + \log(\bar{u}_j - u_{k,j})]. \quad (6)$$

The barrier-augmented running cost is

$$\ell_b(x_k, u_k) = \ell(x_k, u_k) + b_x(x_k; \mu) + b_u(u_k; \sigma). \quad (7)$$

The terminal state barrier is included whenever x_N is constrained. Because the logarithm is defined only in the strict interior, every accepted Box-iLQR iterate remains strictly feasible with respect to the discrete box constraints. This differs from penalty and augmented-Lagrangian approaches, which can admit infeasible intermediate iterates while driving the constraint residual toward zero. Thus, feasibility is enforced directly by the domain of the optimization problem rather than only through a penalty on constraint violation.

For a constrained scalar control u_j ,

$$(b_u)_{u_j} = -\sigma_j \left[\frac{1}{u_j - \underline{u}_j} - \frac{1}{\bar{u}_j - u_j} \right], \quad (8)$$

$$(b_u)_{u_j u_j} = \sigma_j \left[\frac{1}{(u_j - \underline{u}_j)^2} + \frac{1}{(\bar{u}_j - u_j)^2} \right] > 0. \quad (9)$$

Analogous expressions hold for constrained state components. The barrier therefore contributes both a repulsive gradient and a rapidly increasing positive curvature near each boundary, as in standard logarithmic interior-point methods [20].

III. BOX-iLQR

Box-iLQR retains the standard backward–forward structure of iLQR while incorporating the state and control barriers defined in Sec. II. As summarized in Algorithm 1, each iteration first performs a backward pass along the current nominal trajectory to compute a local feedforward correction and feedback policy. A nonlinear forward rollout is then performed using this policy, with line search used to ensure sufficient decrease and strict feasibility. Previous work provides the convergence analysis and associated theoretical guarantees for the underlying log-barrier formulation [21].

Let $(\bar{x}_k, \bar{u}_k)$ denote the current nominal trajectory. Linearizing the dynamics about this trajectory gives

$$\delta x_{k+1} = A_k \delta x_k + B_k \delta u_k, \quad (10)$$

where $A_k = f_x(\bar{x}_k, \bar{u}_k)$, and $B_k = f_u(\bar{x}_k, \bar{u}_k)$.

During the *backward pass* of Algorithm 1, the barrier-augmented stage cost is quadratically approximated along the nominal trajectory. Combining this approximation with the value function at $k+1$ gives

$$Q_x = \ell_{b,x} + A_k^\top V_x^{k+1}, \quad Q_u = \ell_{b,u} + B_k^\top V_x^{k+1}, \quad (11)$$

$$Q_{xx} = \ell_{b,xx} + A_k^\top V_{xx}^{k+1} A_k, \quad Q_{uu} = \ell_{b,uu} + B_k^\top V_{xx}^{k+1} B_k, \quad (12)$$

$$Q_{ux} = \ell_{b,ux} + B_k^\top V_{xx}^{k+1} A_k. \quad (13)$$

The terminal conditions on V_x and V_{xx} are

$$V_x^N = \phi_x, \quad \text{and} \quad V_{xx}^N = \phi_{xx}.$$

When required, the control Hessian is regularized as

$$\tilde{Q}_{uu} = Q_{uu} + \lambda I, \quad \lambda \geq 0, \quad (14)$$

so that it remains positive definite.

Minimization of the local quadratic model gives the affine control correction

$$\delta u_k = k_k + K_k \delta x_k, \quad (15)$$

where

$$k_k = -\tilde{Q}_{uu}^{-1} Q_u, \quad K_k = -\tilde{Q}_{uu}^{-1} Q_{ux}. \quad (16)$$

The value-function derivatives are then propagated backward in time using the usual Riccati recursion. Because the barrier

derivatives are included in ℓ_b , both state and control constraints directly modify the same backward recursion that produces k_k and K_k .

During the *forward pass* of Algorithm 1, the nonlinear system is rolled out using

$$u_k^+(\alpha) = \bar{u}_k + \alpha k_k + K_k(x_k^+ - \bar{x}_k), \quad (17)$$

$$x_{k+1}^+ = f(x_k^+, u_k^+), \quad x_0^+ = x_0. \quad (18)$$

Here $\alpha \in (0, 1]$ scales the feedforward update while retaining the feedback correction. Before the barrier cost is evaluated, the complete nonlinear rollout is checked for strict satisfaction of all box constraints. If the rollout is infeasible or does not provide sufficient decrease, Algorithm 1 reduces the step according to

$$\alpha \leftarrow \beta_\alpha \alpha, \quad 0 < \beta_\alpha < 1. \quad (19)$$

Only when feasibility cannot be recovered for $\alpha \geq \bar{\alpha}$ is the corresponding barrier parameter increased and the backward pass recomputed. Thus, the trajectory update is first globalized through line search before the barrier model itself is modified. This ordering is motivated by the infeasible-Newton behavior of pure primal barrier methods [19].

Full DDP additionally includes contractions between the value gradient and second derivatives of the dynamics. These terms can improve local convergence near a solution, but away from the optimum they need not preserve positive curvature and can produce a poor local model. Recent work shows that iLQR corresponds to a consistent SQP approximation of the discrete optimal-control problem and retains stronger descent properties [5], [6]. Since barrier continuation repeatedly changes the local objective, we use the more robust iLQR approximation.

Algorithm 1 summarizes the feasibility-first continuation strategy. If no feasible rollout is recovered, the barrier associated with the violated constraint is increased; if feasible rollouts exist but none provides sufficient decrease, the regularization is increased instead. After convergence of the fixed-barrier subproblem, the barrier parameters are reduced toward γ_{stop} . For nominal constrained optimization, γ_{stop} is chosen near zero, whereas for feedback deployment continuation may terminate at a finite $\gamma_{\text{fb}} > 0$.

IV. FINITE BARRIERS AS FEEDBACK SHAPING

The logarithmic barrier does more than enforce feasibility during trajectory optimization: its gradient modifies the nominal trajectory, while its curvature directly affects the feedback gains computed in the backward pass.

Consider a scalar constrained quantity $z \in (\underline{z}, \bar{z})$ with barrier

$$b(z; \gamma) = -\gamma [\log(z - \underline{z}) + \log(\bar{z} - z)]. \quad (20)$$

Its descent contribution is

$$-\nabla b = \gamma \left[\frac{1}{z - \underline{z}} - \frac{1}{\bar{z} - z} \right], \quad (21)$$

Algorithm 1 Box-iLQR

Require: Strictly feasible $(\bar{x}, \bar{u})$, barriers $\gamma = [\mu^\top, \sigma^\top]^\top > 0$

- 1: **while** termination condition not satisfied **do**
- 2: Linearize dynamics and quadratize the barrier-augmented cost
- 3: Backward pass: compute $(k_k, K_k)_{k=0}^{N-1}$ using (11)–(16)
- 4: $\alpha \leftarrow 1$, accepted $\leftarrow$ false
- 5: **while** $\alpha \geq \bar{\alpha}$ **do**
- 6: Roll out (17)–(18)
- 7: **if** rollout is infeasible **then**
- 8: $\alpha \leftarrow \beta_\alpha \alpha$
- 9: **else if** sufficient decrease is satisfied **then**
- 10: $(\bar{x}, \bar{u}) \leftarrow (x^+, u^+)$
- 11: accepted $\leftarrow$ true
- 12: **break**
- 13: **else**
- 14: $\alpha \leftarrow \beta_\alpha \alpha$
- 15: **end if**
- 16: **end while**
- 17: **if** no feasible rollout was found **then**
- 18: Increase offending barrier parameter(s)
- 19: **else if not** accepted **then**
- 20: Increase regularization λ
- 21: **else if** fixed-barrier problem has converged **then**
- 22: Reduce γ toward γ_{stop}
- 23: **end if**
- 24: **end while**
- 25: **return** (x^*, u^*, K^*)

which increasingly repels the nominal trajectory from a nearby boundary. Along the central path, perturbed complementarity gives approximately

$$s \approx \frac{\gamma}{\lambda^*}, \quad (22)$$

where s is the active constraint slack and λ^* is the corresponding Lagrange multiplier [20] for hard constraint. Hence, retaining $\gamma > 0$ deliberately preserves finite interior margin.

For a constrained input component $u_{k,j}$, write the control Hessian as

$$\tilde{Q}_{uu,k} = H_k + D_{b,k}, \quad (23)$$

where H_k contains the non-barrier terms, and $D_{b,k}$ is the diagonal barrier Hessian. Its j th diagonal entry is

$$[D_{b,k}]_{jj} = \sigma_j \left[\frac{1}{(u_{k,j} - \underline{u}_j)^2} + \frac{1}{(\bar{u}_j - u_{k,j})^2} \right]. \quad (24)$$

Proposition 1. Suppose H_k and $Q_{ux,k}$ remain bounded and the block of $\tilde{Q}_{uu,k}$ associated with the remaining controls remains uniformly nonsingular. For fixed $\sigma_j > 0$, if $u_{k,j}$ approaches either of its bounds, then

$$K_{k,j} \rightarrow 0. \quad (25)$$

Proof: Let

$$d_j = [D_{b,k}]_{jj}. \quad (26)$$

From (24), $d_j \rightarrow \infty$ as the corresponding slack tends to zero. After permuting the j th control to the first coordinate,

$$\tilde{Q}_{uu,k} = \begin{bmatrix} d_j + \mathcal{O}(1) & b^\top \\ b & C \end{bmatrix}, \quad (27)$$

where b and C^{-1} remain bounded. The block inverse formula therefore gives

$$e_j^\top \tilde{Q}_{uu,k}^{-1} = \mathcal{O}(d_j^{-1}). \quad (28)$$

Since $Q_{ux,k}$ is bounded,

$$K_{k,j} = -e_j^\top \tilde{Q}_{uu,k}^{-1} Q_{ux,k} \rightarrow 0. \quad (29)$$

For a slack s_j from the approaching bound,

$$[D_{b,k}]_{jj} \approx \left(\frac{\sigma_j}{s_j^2} \right), \quad (30)$$

so Proposition 1 gives

$$\|K_{k,j}\| = \mathcal{O}\left(\frac{s_j^2}{\sigma_j}\right), \quad s_j \rightarrow 0. \quad (31)$$

CL-DDP exhibits a related limiting behavior through its active-set formulation. When a control is clamped at a bound, the corresponding feedback row is set to zero [9]. Box-iLQR reaches the same limiting structure continuously: as the nominal input approaches saturation, the logarithmic barrier curvature increases and progressively attenuates the corresponding feedback gain. Thus, CL-DDP changes the local feedback structure through a discrete active-set decision, whereas Box-iLQR modifies it smoothly through the barrier Hessian.

State barriers influence the feedback policy through the same backward pass. Their curvature enters $\ell_{b,xx}$ and is propagated backward through the value Hessian V_{xx} . At an earlier time step,

$$Q_{uu} = \ell_{b,uu} + B^\top V_{xx}^{k+1} B, \quad (32)$$

$$Q_{ux} = \ell_{b,ux} + B^\top V_{xx}^{k+1} A, \quad (33)$$

and therefore

$$K = -Q_{uu}^{-1} Q_{ux}. \quad (34)$$

Consequently, curvature generated by a future state constraint influences earlier feedback gains before the nominal trajectory reaches the constrained region. We refer to this effect as constraint-aware or anticipatory feedback.

Finally, we distinguish between a barrier used to approximate the hard-constrained optimum and one retained for feedback deployment:

$$(x_{\text{opt}}, u_{\text{opt}}, K_{\text{opt}}) : \gamma_{\text{stop}} = \epsilon_{\text{opt}}, \quad \epsilon_{\text{opt}} \ll 1, \quad (35)$$

$$(x_{\text{fb}}, u_{\text{fb}}, K_{\text{fb}}) : \gamma_{\text{stop}} = \gamma_{\text{fb}} > 0. \quad (36)$$

The first provides a closer approximation to the hard-constrained nominal solution. The second intentionally retains state and actuator margin and is used when the feedback

policy itself is an important output of the optimization. The finite barrier therefore trades nominal optimality for interior margin and potentially improved closed-loop robustness.

Although we do not provide guarantees of constraint satisfaction under uncertainty, we evaluate its closed-loop effect empirically in Sect. V.

V. EXPERIMENTS

Box-iLQR is evaluated on pendulum, cart-pole, Acrobot, and car-parking benchmarks and on a data-driven MuJoCo fish model. For the baseline comparisons, ALTRO is implemented using the open-source `Altro.jl` package [14], while CL-DDP is implemented following the control-limited DDP formulation of Tassa *et al.* [9]. Reported iteration counts are solver-specific convergence statistics and should not be interpreted as direct computational-time comparisons.

The pendulum, cart-pole, and Acrobot controls satisfy $|u| \leq 1$, $|u| \leq 2$, and $|u| \leq 5$, respectively. For the state-constrained cart-pole, the cart position additionally satisfies $|x| \leq 0.15$ m. The car has componentwise control bounds $|u_1| \leq 0.5$ and $|u_2| \leq 2$; its state-constrained variant additionally requires $|x|, |y| \leq 1.5$ m. The control-only car-parking task uses $x_g = [0, 0, 0, 0]^T$ as in [9], whereas the state-constrained task uses $x_g = [0, 0, 2\pi, 0]^T$.

A. Benchmark Comparison

Table I shows that Box-iLQR obtains solutions comparable to the established constrained solvers on the control-limited benchmarks. For the pendulum, Box-iLQR and CL-DDP obtain nearly identical costs, 55.49 and 55.46, respectively. The same behavior is observed for the control-constrained cart-pole, where the corresponding costs are 762.45 and 762.44. For control-only car parking, ALTRO obtains the lowest reported objective, 1.4361, followed by Box-iLQR at 1.9065 and CL-DDP at 1.9448. ALTRO converged to a better solution than Box-iLQR/CL-DDP.

All three methods converge for the Acrobot. ALTRO obtains the lowest cost, 39581.50, followed by Box-iLQR at 40494.38 and CL-DDP at 41712.44. Figure 1 compares the resulting trajectories. The different objective values emphasize that the methods are local nonlinear optimizers and need not converge to identical stationary solutions even from the same task initialization.

The main distinction appears when state constraints are introduced. For the state-constrained cart-pole, ALTRO converges to a feasible solution with higher cost, 912.38, while the CL-DDP formulation considered here does not accommodate state constraints. Box-iLQR converges with cost 765.05. For state-constrained car parking, both ALTRO and Box-iLQR converge to constraint-feasible trajectories, while the CL-DDP formulation considered here is not applicable. As shown in Fig. 2, the ALTRO and Box-iLQR reach the desired terminal configuration $x_g = [0, 0, 2\pi, 0]^T$, with a final cost of 0.70 and 0.69, respectively. For the state-constrained car-parking task, ALTRO was sensitive to initialization and solver parameters in our experiments; the solution reported

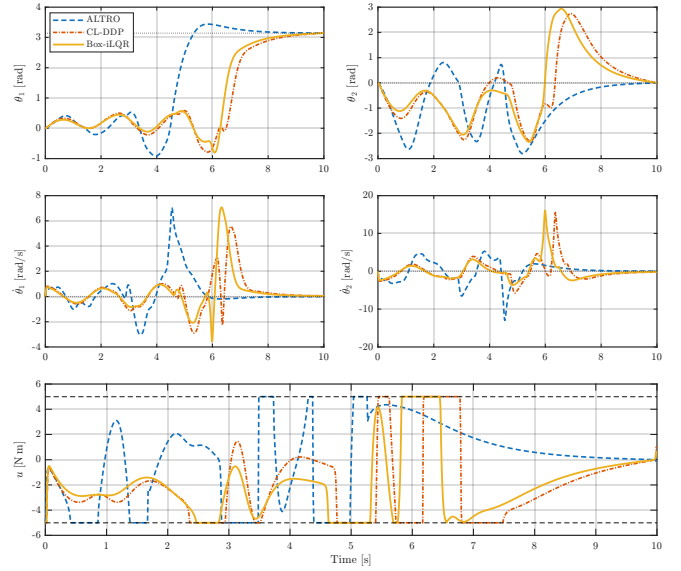


Fig. 1. Acrobot swing-up using ALTRO, CL-DDP, and Box-iLQR. All three solutions satisfy the actuator constraint $|u| \leq 5$.

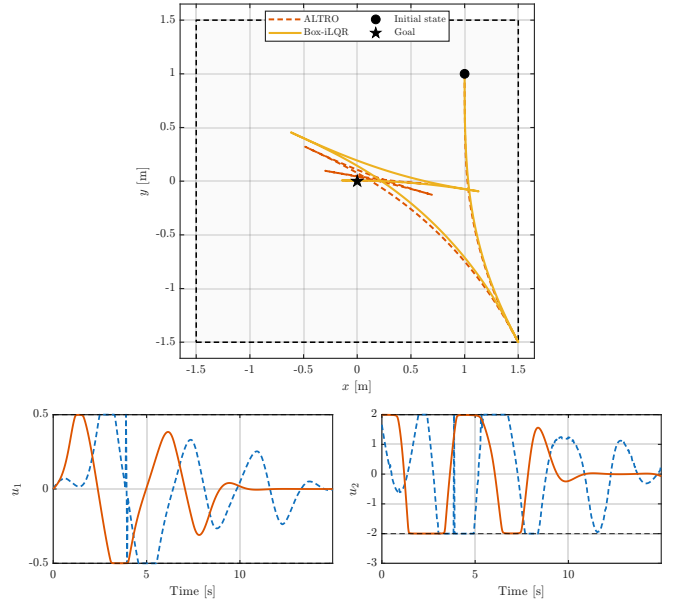


Fig. 2. State- and control-constrained car parking. Both trajectories satisfy the position constraints $-1.5 \leq x, y \leq 1.5$. Both Box-iLQR and ALTRO reach the desired terminal configuration $x_g = [0, 0, 2\pi, 0]^T$, with Box-iLQR attaining a slightly lower nominal cost. The lower panels show the corresponding control histories and bounds.

in Table I is the converged constraint-feasible result used for comparison.

The closed-loop effect of the resulting feedback law is examined later in Fig. 5, where Box-iLQR feedback is shown to eliminate the state-constraint violations observed under open-loop execution in the tested Monte Carlo set.

These results indicate that Box-iLQR is not uniformly superior in iteration count or nominal objective value. Its main advantage is instead the ability to treat state and control box constraints in the same Riccati-based formulation while retaining a local feedback policy.

TABLE I

BENCHMARK COMPARISON. COSTS ARE EVALUATED USING THE COMMON NOMINAL OBJECTIVE AND EXCLUDE BARRIER, PENALTY, AND AUGMENTED-LAGRANGIAN TERMS. (N/A - CL-DDP CANNOT ACCOMMODATE STATE CONSTRAINTS.)

Problem	Constraints	ALTRO		CL-DDP		Box-iLQR	
		Iterations	Cost	Iterations	Cost	Iterations	Cost
Pendulum	Control	436	67.24	206	55.46	527	55.49
Cart-pole	Control	1091	994.92	3554	762.44	8226	762.45
Cart-pole	State + control	968	912.38	N/A	—	3729	765.05
Acrobot	Control	4665	39581.50	4797	41712.44	4096	40494.38
Car parking	Control	142	1.4361	1597	1.9448	335	1.9065
Car parking	State + control	113	0.70	N/A	—	954	0.69

Box-iLQR maintains strict feasibility of every accepted iterate, whereas the augmented-Lagrangian mechanism used by ALTRO does not require intermediate trajectories to remain feasible. This property is particularly useful when intermediate trajectories are passed to a simulator, hardware system, or another planning layer that assumes constraint-feasible states and controls.

B. Finite-Barrier Feedback

We next examine whether the barrier value that best approximates the hard-constrained optimum also provides the most useful feedback controller. For stochastic execution, the applied input is

$$u_k^{(q)} = \text{sat}_{[\underline{u}, \bar{u}]} \left(u_{k,\text{fb}}^{(q)} + \rho |u_{k,\text{fb}}^{(q)}| \odot \xi_k^{(q)} \right), \quad (37)$$

where $u_{k,\text{fb}}^{(q)} = u_k^* + K_k(x_k^{(q)} - x_k^*)$, $\xi_k^{(q)} \sim \mathcal{N}(0, I)$. Thus, the disturbance magnitude scales with the current feedback-corrected input.

For the control-constrained cart-pole, feedback obtained at $\sigma \simeq 0.03$ is compared with Box-iLQR continued to approximately $\sigma = 10^{-6}$ and with CL-DDP. As shown in Fig. 3, the finite-barrier feedback produces substantially smaller terminal errors ($\|x_N - x_g\|_2$) over the tested uncertainty range than the aggressively continued Box-iLQR solution and CL-DDP. The finite-barrier nominal solution still reaches the upright equilibrium with terminal error of order 10^{-3} in 1000 Monte Carlo realizations. Using the feedback, the terminal error is almost zero for feedback at barrier parameter ≈ 0.03 . The shaded region in Fig. 3 denotes $\bar{e}_N \pm 3s_e$, where s_e is the sample standard deviation of the terminal error. In addition to achieving a lower mean terminal error, the finite-barrier Box-iLQR policy exhibits a narrow $3s_e$ envelope, indicating low variability across the Monte Carlo realizations. Thus, its improved closed-loop performance is not limited to the mean response; the terminal errors remain tightly concentrated across the tested realizations.

For the state-constrained cart-pole, we retain the larger barrier $\gamma_{\text{fb}} \approx 0.02$ to provide additional interior margin and perform 100 closed-loop simulations at $\rho = 0.05$. Figure 4 shows all four states and the applied control. No cart-position constraint violation is observed in the tested realizations. The state trajectories have very small dispersion and therefore

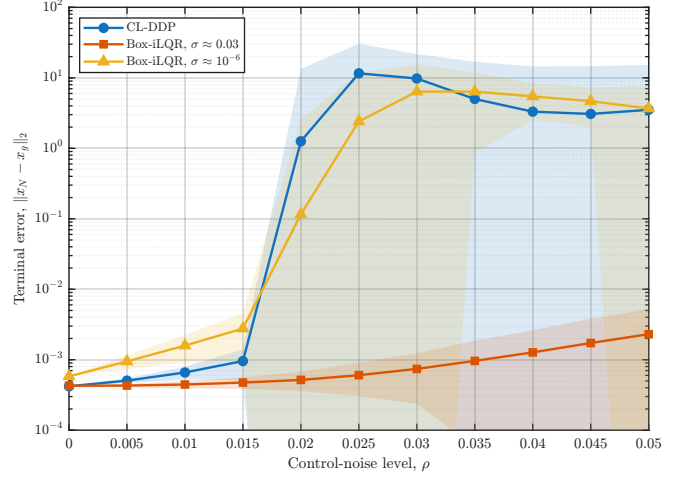


Fig. 3. Cart-pole terminal error under execution uncertainty. Solid curves show the mean over 1000 Monte Carlo realizations, and the shaded regions denote $\bar{e}_N \pm 3s_e$, where s_e is the sample standard deviation of the terminal error. Box-iLQR feedback at $\sigma \simeq 0.03$ is compared with Box-iLQR at $\sigma \simeq 10^{-6}$ and CL-DDP.

largely overlap the nominal trajectory; the individual Monte Carlo realizations are more readily visible in the control history.

We perform the same closed-loop evaluation for the state-constrained car-parking problem. Figure 5 shows 100 Monte Carlo realizations at a finite barrier value of approximately $\sigma \approx 0.02$ and noise level $\rho = 0.05$. The controls are saturated at their prescribed bounds, $|u_1| \leq 0.5$ and $|u_2| \leq 2$, while the position constraints are $|x|, |y| \leq 1.5$ m.

We compare open-loop execution, $u_k = u_k^*$, with the Box-iLQR feedback policy, $u_k = u_k^* + K_k(x_k - x_k^*)$, under identical realizations of the multiplicative execution uncertainty in (37). Table II summarizes the results. For the car-parking problem, 29 of 100 open-loop realizations violate the state constraints, whereas no violation is observed among the 100 realizations using Box-iLQR feedback. Box-iLQR likewise achieves a 100% success rate for the state-constrained cart-pole problem. These results demonstrate that the feedback substantially improves robustness across both state-constrained examples. A closed-loop comparison with ALTRO is not included because ALTRO is used typically to compute constrained nominal trajectories and does not

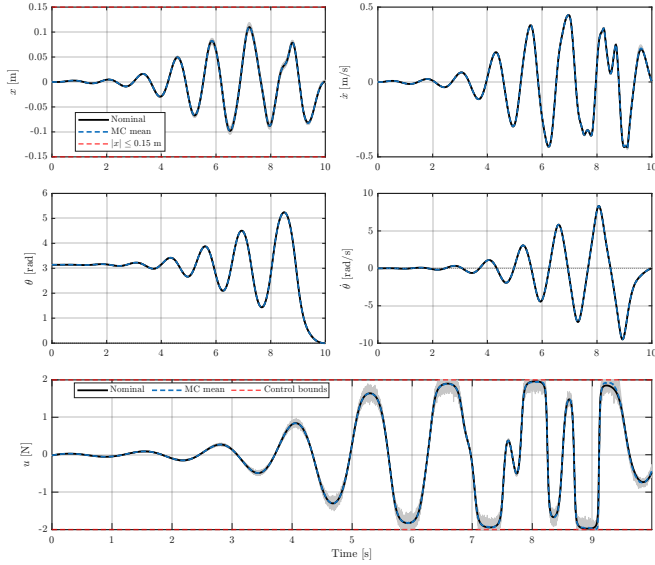


Fig. 4. State-constrained cart-pole feedback at $\sigma \simeq \mu \simeq 0.02$ and $\rho = 0.05$. Gray curves denote 100 Monte Carlo realizations. The constraints are $|x| \leq 0.15$ m and $|u| \leq 2$; no state violation is observed in the tested set.

TABLE II

STATE-CONSTRAINT VIOLATIONS OVER 100 MONTE CARLO RUNS.

Problem	Open loop	Box-iLQR (finite-barrier)
Car parking (state)	29/100	0/100
Cart-pole (state)	100/100	0/100

directly provide a feedback policy that can ensure constraint satisfaction under uncertainty as analyzed in this section. In addition, its augmented-Lagrangian iterations need not remain feasible at intermediate steps, unlike the accepted iterates of Box-iLQR, thereby not allowing to use a feedback associated with a sub-optimal iterate as used by Box-iLQR.

C. Data-Driven MuJoCo Fish

The final experiment applies Box-iLQR to the six-input MuJoCo fish [22], with $|u_j| \leq 1$, $j = 1, \dots, 6$. Unlike the analytical benchmarks, the local dynamics are identified from black-box simulator rollouts following the D2C/search-for-feedback viewpoint [7], [8]. Around the nominal trajectory, the linear perturbation model

$$\delta x_{k+1} \approx \hat{A}_k \delta x_k + \hat{B}_k \delta u_k \quad (38)$$

is estimated by least squares from 200 symmetric perturbation rollouts. The identified matrices replace the analytical Jacobians in the Box-iLQR backward pass. The task is to drive the fish toward $p_g = [0 \ 0.4 \ 0.2]^\top$.

Both iLQR and Box-iLQR drove the fish to the target. Figure 6 compares the resulting Box-iLQR control histories with unconstrained iLQR. The unconstrained solution requests inputs outside the admissible interval, whereas data-driven Box-iLQR keeps all six controls within their prescribed bounds. This demonstrates that the same barrier formulation can handle multiple simultaneously constrained actuators even when the local plant Jacobians are obtained entirely from rollout data.

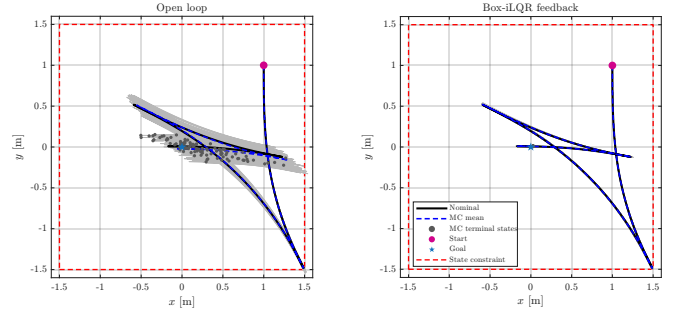


Fig. 5. State-constrained car parking under identical execution disturbances with $\rho = 0.05$. Gray curves show 100 Monte Carlo realizations, with the nominal trajectory in solid black and the Monte Carlo mean dashed. The red boundary denotes $|x|, |y| \leq 1.5$ m. Open-loop execution violates the state constraints in 29/100 runs, while no violation is observed with finite-barrier Box-iLQR feedback at $\gamma_{fb} \approx 0.02$.

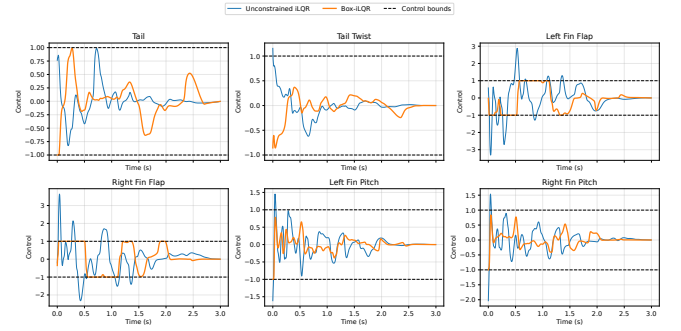


Fig. 6. Six-input MuJoCo fish controls. Unconstrained iLQR exceeds the actuator limits, whereas data-driven Box-iLQR satisfies all six bounds.

Closed-loop robustness is evaluated for $\rho \in \{0.01, 0.02, \dots, 0.50\}$ using 1000 realizations per noise level. To quantify the deviation caused solely by execution uncertainty, we measure the terminal position error relative to the nominal Box-iLQR terminal position: $e_N^{(q)} = \|p_f^{(q)} - p_f^{\text{nom}}\|_2$, where $p_f^{(q)}$ is the terminal fish position in the q th Monte Carlo realization and p_f^{nom} is the nominal terminal position. Figure 7 reports the mean terminal error and $3s_e$ dispersion, where s_e denotes the standard deviation. The results show that finite-barrier feedback recovered from identified local dynamics remains effective across the tested noise range, demonstrating that Box-iLQR can combine constrained trajectory optimization and local feedback recovery without analytical plant Jacobians. Even at $\rho = 0.3$, the $\bar{e}_N \pm 3s_e$ envelope remains below 0.05, indicating tightly concentrated terminal position errors despite 30% multiplicative control noise.

VI. CONCLUSION

This paper presented Box-iLQR, a log-barrier iLQR method for box-constrained states and controls that preserves the standard backward-forward structure and local affine feedback policy of unconstrained iLQR. iLQR is chosen over full DDP because its local approximation provides more reliable descent behavior away from the optimum [6].

The barrier admits a useful feedback interpretation. As a constrained control approaches saturation, its curvature

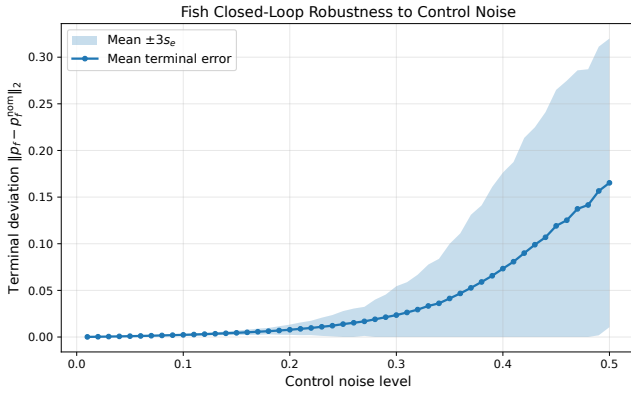


Fig. 7. MuJoCo fish terminal position error relative to the nominal terminal position. The solid curve shows the mean over 1000 Monte Carlo realizations and the shaded region denotes $\bar{e}_N \pm 3s_e$, where s_e is the sample standard deviation of the terminal position error.

increases and smoothly attenuates the corresponding feedback row, while state-barrier curvature propagates backward through the value function and influences feedback before a future state boundary is encountered. This motivates retaining a finite barrier when the feedback law, rather than only the nominal constrained optimum, is the desired output. Unlike penalty-based constrained trajectory optimization, the log-barrier formulation maintains strict feasibility of accepted iterates, while still returning the local feedback policy associated with the constrained trajectory.

Across pendulum, cart-pole, Acrobot, and car-parking benchmarks, Box-iLQR produces competitive control-constrained solutions and directly handles the state-constrained cases considered here. For state-constrained car parking, both ALTRO and Box-iLQR reach the desired terminal configuration, with comparable nominal costs. Closed-loop cart-pole and car-parking experiments demonstrate the value of the barrier-modified feedback policy under execution uncertainty. Retaining a finite barrier improves closed-loop behavior in the cart-pole and car-parking example, while Box-iLQR feedback eliminates the state-constraint violations observed under open-loop car-parking execution in the tested Monte Carlo set. The six-input MuJoCo fish experiment further shows that the same constrained feedback framework can use local dynamics identified entirely from simulator rollouts.

Future work will focus on automatically selecting finite barrier values using disturbance or covariance information, developing infeasible-start and receding-horizon variants, and validating the method on physical robotic platforms. We also plan to extend the feedback framework beyond box constraints to more general state and control constraints.

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